

HYDROGENATED NBR (HNBR) SYNTHESIS AND PROCESS LANDSCAPE FROM PATENT EVIDENCE

1. ABSTRACT

This report synthesizes patent evidence relevant to hydrogenated nitrile butadiene rubber (HNBR), focusing on precursor NBR composition, hydrogenation routes, and process strategies that improve molecular-weight control, residual unsaturation, and manufacturability. Across the selected patents, the target material is consistently defined as a nitrile rubber derived from at least one conjugated diene and at least one α,β -unsaturated nitrile, typically acrylonitrile/butadiene copolymers, which are then hydrogenated partially or selectively to reduce residual double bonds and improve heat and oil resistance. The landscape shows two major synthesis themes: (1) direct hydrogenation of NBR in solution or in specialized reactors using homogeneous catalysts, and (2) precursor engineering via solution polymerization, RAFT-mediated polymerization, or functionalized NBR architectures that are subsequently hydrogenated or coupled.

The hydrogenation patents emphasize catalyst choice and reactor/process intensification. US7666950B2 discloses hydrogenation in a spinning disk reactor with hydrogen pressure from 3.5 to 100 bar and broad catalyst families including rhodium- and ruthenium-based systems. EP2473280B1 and US9150669B2 focus on simultaneous hydrogenation/metathesis using ruthenium or osmium catalysts, enabling lower molecular weight HNBR with narrower molecular-weight distributions and low residual double bond content in a single step. These disclosures highlight the tradeoff between hydrogenation efficiency, metathesis-driven chain scission, and viscosity control. In contrast, EP2478021B1 and EP2554558A1 address the upstream NBR synthesis problem by solution polymerization in organic solvents or solvent mixtures, using controlled radical polymerization concepts and chain-transfer/modifier systems to tune molecular weight before hydrogenation. EP2623518A1 and US20140357801A1 extend this precursor-engineering approach by introducing alkyne- or tetrazole-functional RAFT-derived NBRs that can be post-coupled, with hydrogenation remaining an optional downstream step.

Overall, the evidence indicates that the most robust HNBR manufacturing strategies combine controlled NBR precursor synthesis, careful management of molecular weight and viscosity, and hydrogenation under catalysts/processes that preserve polymer integrity while achieving low residual unsaturation.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7666950B2
- Patent Title: Process for preparing hydrogenated nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparing hydrogenated nitrile rubber by subjecting nitrile rubber to hydrogenation in a spinning disk reactor. The process is intended to reduce the hydrogen pressure burden relative to conventional batch hydrogenation while using homogeneous or heterogeneous hydrogenation catalysts, including rhodium- or ruthenium-based systems. - Hydrogen pressure: 3.5 bar to 100 bar — Claim 1 - Hydrogen pressure: 65 bar to 90 bar — Background discussion of batch hydrogenation - Mooney viscosity (HNBR): 55 to 120 ML 1+4 @100°C — Background discussion - Residual double bond (RDB) content: up to 18% — Background discussion - Mooney viscosity (low Mooney HNBR): 2 to 50 ML 1+4 @100°C — Background discussion - Molecular weight Mw (low Mooney HNBR): 30,000 to 250,000 — Background discussion - Polydispersity index: typically less than 3 — Background discussion - Mooney viscosity (NBR): 30 to 90 ML 1+4 @100°C — General description of NBR - Mooney viscosity (NBR): 30 to 70 ML 1+4 @100°C — General description of NBR - Mooney viscosity (NBR): 30 to 50 ML 1+4 @100°C — General description of NBR - Molecular weight Mw (NBR): 200,000 to 500,000 — General description of NBR - Molecular weight Mn (NBR): 200,000 to 400,000 — General description of NBR - Molecular weight Mn (NBR): 80,000 to 150,000 — General description of NBR - Polydispersity index PDI (NBR): 2.0 to 6.0 — General description of NBR - Polydispersity index PDI (NBR): 2.0 to 4.0 — General description of NBR - Conjugated diene content: 40 to 90 wt% — General description of NBR - Conjugated diene content: 60 to 85 wt% — General description of NBR - α,β -unsaturated nitrile content: 10 to 60 wt% — General description of NBR - α,β -unsaturated nitrile content: 15 to 40 wt% — General description of NBR - Additional copolymerizable monomers: 0 to 40 wt% — General description of NBR - Additional copolymerizable monomers: 0.1 to 40 wt% — General description of NBR - Additional copolymerizable monomers: 1 to 30 wt% — General description of NBR - Polymerization temperature: 100°C — Commercial NBR preparation description - Polymerization time: 9 hours — Example-like process description in detailed text - NBR feed pressure: 3 bar — Example-like process description in detailed text

Relevant Experimental Evidence - Example-like process: NBR precursor was polymerized from acrylonitrile and 1,3-butadiene in DMAc with Vazo 88 and DoPAT, then isolated by ethanol precipitation. - Example-like process: Hydrogenation is described as being carried out in a

spinning disk reactor, with the patent emphasizing lower pressure demand than conventional batch systems. - Example-like process: The patent frames HNBR as a selective hydrogenation product with improved heat, ozone, chemical, and oil resistance.

Technical Relevance This patent is directly relevant because it addresses the core HNBR transformation: hydrogenation of NBR to reduce residual unsaturation. It is especially important for process intensification, showing that reactor design and hydrogen pressure can be optimized for HNBR manufacture.

Patent 2

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of nitrile rubbers in organic solvents using a free-radical process with a modifier, followed optionally by hydrogenation. The emphasis is on producing NBR precursors in organic medium that can later be hydrogenated to HNBR, avoiding some limitations of emulsion-derived latex isolation. - Monomer ratio (Example 1): acrylonitrile 37 phm, 1,3-butadiene 63 phm — Example 1 - Initiator amount (Example 1): Vazo 88 0.31 phm — Example 1 - Modifier amount (Example 1): DoPAT 0.22 phm — Example 1 - Solvent amount (Example 1): dimethylacetamide 51 phm — Example 1 - Acrylonitrile amount (Example 1): 43 mL, 653.3 mmol — Example 1 - 1,3-Butadiene amount (Example 1): 90 mL, 1077 mmol — Example 1 - Hydrogen pressure for conventional batch hydrogenation: 65 to 90 bar — Background discussion - Residual double bond (RDB) content of commercially available HNBR: up to 18% — Background discussion - Mooney viscosity (commercial HNBR): 55 to 120 ML 1+4 @100°C — Background discussion - Mooney viscosity (low Mooney HNBR): 2 to 50 ML 1+4 @100°C — Background discussion - Molecular weight Mw (low Mooney HNBR): 30,000 to 250,000 — Background discussion - Polymerization temperature: 100°C — Example 1 - Polymerization time: 9 hours — Example 1 - Polymerization pressure for butadiene feed: 3 bar — Example 1 - NBR composition range: 40 to 90 wt% diene — General description - NBR composition range: 10 to 60 wt% α,β -unsaturated nitrile — General description - Additional copolymerizable monomers: 0 to 40 wt% — General description

Relevant Experimental Evidence - Example 1: NBR was prepared in DMAc with Vazo 88 and DoPAT, using acrylonitrile and 1,3-butadiene at 100°C and 3 bar feed pressure. - Example 6: A terpolymer containing hydroxyethyl acrylate was prepared by the same general solution route and stopped after 6 hours at 39% conversion. - Example 7: A low-molecular-weight NBR was intentionally prepared by increasing regulator amount to 2.22 phm. - Example 8: The low-MW NBR

from Example 7 was chain-extended with styrene at 100°C for 13 hours. - Example 17: NBRs were prepared with additional HDODA and/or HEA, with polymerization at 100°C and butadiene feed at 3 bar, followed by ethanol precipitation.

Technical Relevance This patent is relevant because it provides a precursor-synthesis route that can be paired with later hydrogenation to HNBR. It is useful for controlling NBR molecular weight and functionality before hydrogenation, which can influence final HNBR processability and properties.

Patent 3

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber using ruthenium- or osmium-based catalysts in the presence of hydrogen, optionally with a co-olefin. The process is designed to produce low-molecular-weight HNBR with narrower molecular-weight distribution and low residual double bond content in a single step. - Catalyst amount (HNBR 1): 0.057 phr — Example 1 - Catalyst amount (HNBR 2): 0.080 phr — Example 1 - Catalyst amount (HNBR 3): 0.06 phr — Example 1 - Catalyst amount (HNBR 4): 0.055 phr — Example 1 - Metal content (HNBR 1): 65.6 ppm — Example 1 - Metal content (HNBR 2): 91.3 ppm — Example 1 - Metal content (HNBR 3): 65.6 ppm — Example 1 - Metal content (HNBR 4): 65.6 ppm — Example 1 - Reaction time: 0, 60, 120, 180, 240 min — Example 1 Table 2 - Residual double bond (RDB): 100%, 2.0%, 0.4%, 0.3% — HNBR 1 profile, Example 1 - Residual double bond (RDB): 100%, 0.7%, 0.5%, 0.3% — HNBR 2 profile, Example 1 - Residual double bond (RDB): 100%, 15.4%, 3.1%, 1.3%, 0.6% — HNBR 3 profile, Example 1 - Residual double bond (RDB): 100%, 2.9%, 0.6%, 0.3% — HNBR 4 profile, Example 1 - Mw (HNBR 1): 191541 g/mol — Example 1 Table 3 - Mn (HNBR 1): 68643 g/mol — Example 1 Table 3 - PDI (HNBR 1): 2.79 — Example 1 Table 3 - Mooney viscosity (HNBR 1): 38.2 — Example 1 Table 3 - Mw (HNBR 2): 135772 g/mol — Example 1 Table 3 - Mn (HNBR 2): 58259 g/mol — Example 1 Table 3 - PDI (HNBR 2): 2.33 — Example 1 Table 3 - Mooney viscosity (HNBR 2): 20.2 — Example 1 Table 3 - Mw (HNBR 3): 195304 g/mol — Example 1 Table 3 - Mn (HNBR 3): 64130 g/mol — Example 1 Table 3 - PDI (HNBR 3): 3.05 — Example 1 Table 3 - Mooney viscosity (HNBR 3): 69.3 — Example 1 Table 3 - Mw (HNBR 4): 144212 g/mol — Example 1 Table 3 - Mn (HNBR 4): 59851 g/mol — Example 1 Table 3 - PDI (HNBR 4): 2.45 — Example 1 Table 3 - Mooney viscosity (HNBR 4): 32.6 — Example 1 Table 3

Relevant Experimental Evidence - Example 1: Four catalyst systems were compared; Grubbs III, Grubbs II, and Wilkinson's catalyst were benchmarked for hydrogenation/metathesis performance. - Example 1: HNBR 1 and HNBR 2 showed faster RDB reduction than Wilkinson's catalyst under comparable metal loading. - Example 1: The resulting HNBRs had Mw values from 135772 to 195304 g/mol and PDI values from 2.33 to 3.05. - Example 1: HNBR 2 gave the lowest Mooney viscosity at 20.2, while HNBR 3 gave the highest at 69.3. - Example 1: The patent demonstrates that catalyst identity strongly affects hydrogenation rate and final polymer properties.

Technical Relevance This patent is highly relevant because it directly targets HNBR formation via simultaneous hydrogenation and chain scission/metathesis. It provides concrete catalyst and property data showing how to tune residual unsaturation, molecular weight, and viscosity in HNBR.

Patent 4

- Patent Number: EP2623518A1
- Patent Title: Nitrile rubbers coupled with bistriazolyl groups, production and use of same
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of alkyne-functional NBR by RAFT-mediated solution polymerization, followed by optional coupling through click-type chemistry. Hydrogenation is described as an allowable downstream modification of the NBR precursor, making the patent relevant to functional HNBR precursor design rather than direct hydrogenation chemistry. - DoPAT synthesis: 12.000 g, 34.2 mmol — Example 1 - Propargyl alcohol: 6.276 g, 112.0 mmol — Example 1 - 4-(Dimethylamino)pyridine: 0.834 g, 6.8 mmol — Example 1 - DCC: 7.056 g, 34.2 mmol — Example 1 - Reaction temperature: 0°C then room temperature overnight — Example 1 - Yield of alkyne-RAFT: 8.409 g, 64% — Example 1 - Vazo 88: 78.2 mg, 0.32 mmol — Example 2 - Alkyne-RAFT: 996.1 mg, 2.56 mmol — Example 2 - Monochlorobenzene: 25.0 mL — Example 2 - Acrylonitrile: 21.7 mL, 329.6 mmol — Example 2 - 1,3-Butadiene: 45.0 mL, 540.8 mmol — Example 2 - Polymerization temperature: 100°C — Example 2 - Polymerization time: 5 h — Example 2 - Final conversion: 19.1% — NBR 2, Example 2 - Final conversion: 19.5% — NBR 3, Example 2 - Final conversion: 10.9% — NBR 4, Example 2 - Final conversion: 10.0% — NBR 5, Example 2 - Final conversion: 15.1% — NBR 6, Example 2 - Mn: 1.0 to 69 kg/mol — Example 2 Table 1 - Mw: 1.2 to 108 kg/mol — Example 2 Table 1 - PDI: 1.2 to 1.5 — Example 2 Table 1 - Coupling temperature: 25°C — Example 3 - Coupling time: 16 h — Example 3

Relevant Experimental Evidence - Example 1: An alkyne-functional RAFT chain-transfer agent was synthesized in 64% yield. - Example 2: Alkyne-functional NBRs were prepared in monochlorobenzene at 100°C using Vazo 88 and alkyne-RAFT. - Example 2: The NBR products showed Mn values from 1.0 to 69 kg/mol and narrow PDI values from 1.2 to 1.5. - Example 3: Alkyne-functional NBRs were coupled with 1,4-bis(azidomethyl)benzene at 25°C for 16 h. - Example 3: Coupling increased molecular weight substantially, reaching Mw up to 158 kg/mol.

Technical Relevance This patent is relevant because it demonstrates controlled NBR precursor synthesis with functional handles that can be carried into HNBR-related architectures. While not a direct hydrogenation patent, it informs precursor design and molecular-weight control before hydrogenation.

Patent 5

- Patent Number: EP2554558A1
- Patent Title: Method for producing nitrile rubbers in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses radical polymerization of nitrile rubber in mixed organic solvents, with the resulting NBR optionally subjected to hydrogenation. The process uses at least two solvents, with a main solvent present at 70 to 99.9 vol% of the solvent mixture, to improve polymerization behavior and time-conversion performance. - Main solvent fraction: 70 to 99.9 vol% — Claim 1 - Monomer ratio (Example 1): acrylonitrile 36 phm, 1,3-butadiene 64 phm — Example 1 - Initiator amount (Example 1): Vam 110 0.38 phm — Example 1 - Modifier amount (Example 1): DoPAT 0.086 phm — Example 1 - Monochlorobenzene amount (Example 1): 139 phm — Example 1 - Dimethylacetamide amount (Example 1): 6.6 phm — Example 1 - Acrylonitrile amount (Example 1): 31.7 mL, 481.5 mmol — Example 1 - 1,3-Butadiene amount (Example 1): 70.5 mL, 847.2 mmol — Example 1 - Polymerization temperature: 100°C — Example 1 - Polymerization time: 22 h — Example 1 - NBR composition range: at least one conjugated diene and at least one α,β -unsaturated nitrile — Claim 1 - Hydrogenation is optional after polymerization — Abstract and claims

Relevant Experimental Evidence - Example 1: NBR was prepared in a monochlorobenzene/dimethylacetamide solvent mixture using Vam 110 and DoPAT at 100°C. - Example 1: The polymerization was run for 22 h and the polymer was isolated by ethanol precipitation. - The patent claims improved time-conversion curves from the mixed-solvent approach. - The process is positioned as a precursor route for later hydrogenation to HNBR.

Technical Relevance This patent is relevant because it improves the synthesis of NBR precursors that can be hydrogenated to HNBR. Solvent selection and mixed-solvent control are important for precursor molecular weight, conversion, and downstream hydrogenation behavior.

Patent 6

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber using ruthenium- or osmium-containing catalysts, with the goal of producing low-molecular-weight HNBR in a single step. The process is designed to reduce residual double bond content while also lowering molecular weight and narrowing molecular-weight distribution. - Catalyst amount (HNBR 1): 0.041 phr — Example 1 - Catalyst amount (HNBR 2): 0.045 phr — Example 1 - Catalyst amount (HNBR 3): 0.060 phr — Example 1 - Metal content: 65.6 ppm — Example 1 - Co-catalyst for HNBR 1: NIL — Example 1 - Co-catalyst for HNBR 2: NIL — Example 1 - Co-catalyst for HNBR 3: triphenylphosphine — Example 1 - Co-catalyst amount: 1.0 phr — Example 1 - Reaction time: 0, 60, 120, 180, 240 min — Example 1 Table 2 - Residual double bond (RDB): 100%, 2.5%, 0.6%, 0.5%, 0.4% — HNBR 1 profile, Example 1 - Residual double bond (RDB): 100%, 7.0%, 4.5%, 3.9%, 3.5% — HNBR 2 profile, Example 1 - Residual double bond (RDB): 100%, 15.4%, 3.1%, 1.3%, 0.7% — HNBR 3 profile, Example 1 - Mw (HNBR 1): 113371 g/mol — Example 1 Table 3 - Mn (HNBR 1): 48352 g/mol — Example 1 Table 3 - PDI (HNBR 1): 2.34 — Example 1 Table 3 - Mooney viscosity (HNBR 1): 33.3 — Example 1 Table 3 - Mw (HNBR 2): 133758 g/mol — Example 1 Table 3 - Mn (HNBR 2): 54595 g/mol — Example 1 Table 3 - PDI (HNBR 2): 2.45 — Example 1 Table 3 - Mooney viscosity (HNBR 2): 41.3 — Example 1 Table 3 - Mw (HNBR 3): 195304 g/mol — Example 1 Table 3 - Mn (HNBR 3): 64130 g/mol — Example 1 Table 3 - PDI (HNBR 3): 3.05 — Example 1 Table 3 - Mooney viscosity (HNBR 3): 69.3 — Example 1 Table 3 - Target HNBR Mw range: 20,000 to 250,000 — Description - Target HNBR Mooney viscosity range: 1 to 50 — Description - Target HNBR MWD: less than 3.0 — Description

Relevant Experimental Evidence - Example 1: Three catalyst systems were compared, including Hoveyda, Arlt, and Wilkinson's catalyst. - Example 1: HNBR 1 achieved the fastest reduction in residual double bonds among the tested systems. - Example 1: HNBR 1 and HNBR 2 produced lower molecular weights than HNBR 3. - Example 1: HNBR 1 had Mw 113371 g/mol and Mooney viscosity 33.3. - Example 1: The patent demonstrates that catalyst selection controls both hydrogenation rate and polymer degradation during HNBR formation.

Technical Relevance This patent is directly relevant because it targets the same HNBR transformation and explicitly optimizes for low molecular weight and low residual unsaturation. It provides practical catalyst and property benchmarks for single-step HNBR synthesis.

Patent 7

- Patent Number: US20140357801A1
- Patent Title: Nitrile rubbers coupled via bisdihydropyrazole groups, production thereof and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of nitrile rubbers bearing tetrazole groups by RAFT-mediated solution polymerization, followed by UV-induced coupling with bifunctional ene compounds to form bisdihydropyrazole-linked materials. Hydrogenation is described as an optional downstream step on the tetrazole-functional NBR precursor, making the patent relevant to functional precursor design for HNBR-related materials. - DoPAT: 5.000 g, 14.3 mmol — Example 1 - 1,3-Propanediol: 5.2 mL, 71.3 mmol — Example 1 - 4-(Dimethylamino)pyridine: 0.346 g, 2.8 mmol — Example 1 - DCC: 2.940 g, 2.8 mmol — Example 1 - Hydroxy-RAFT yield: 4.367 g, 74% — Example 1 - Vazo 88: 44.3 mg, 0.13 phm — Example 2 - Tetrazole-RAFT: 953.0 mg, 2.7 phm — Example 2 - Monochlorobenzene: 18.8 mL, 59.1 phm — Example 2 - Acrylonitrile: 16.4 mL, 37.6 phm — Example 2 - 1,3-Butadiene: 33.8 mL, 62.4 phm — Example 2 - Polymerization temperature: 100°C — Example 2 - Polymerization time: 5 h — Example 2 - Final conversion: 18.0% — NBR 1, Example 2 - Final conversion: 18.5% — NBR 2, Example 2 - Final conversion: 10.8% — NBR 3, Example 2 - Final conversion: 8.9% — NBR 4, Example 2 - Final conversion: 12.0% — NBR 5, Example 2 - Final conversion: 37.2% — NBR 6, Example 2 - Mn: 1.2 to 75 kg/mol — Example 2 Table 1 - Mw: 1.4 to 121 kg/mol — Example 2 Table 1 - PDI: 1.2 to 1.6 — Example 2 Table 1 - Coupling temperature: 25°C — Example 3 - Coupling time: 120 to 180 min — Example 3 - 1,6-Bis(maleimido)hexane: 0.33 to 2.10 mg — Example 3 - Tetrazole-NBR charge: 30.0 to 120 mg — Example 3

Relevant Experimental Evidence - Example 1: A tetrazole-functional RAFT chain-transfer agent was synthesized in 74% yield. - Example 2: Tetrazole-functional NBRs were prepared in monochlorobenzene at 100°C using Vazo 88 and tetrazole-RAFT. - Example 2: The NBR products showed Mn values from 1.2 to 75 kg/mol and Mw values from 1.4 to 121 kg/mol. - Example 3: Tetrazole-functional NBRs were coupled with 1,6-bis(maleimido)hexane at 25°C. - Example 3: Coupling times of 120 to 180 min produced higher-molecular-weight coupled polymers.

Technical Relevance This patent is relevant because it shows how RAFT-derived NBR precursors can be functionalized and later modified, with hydrogenation remaining an optional step. It contributes to the broader HNBR precursor engineering space, especially where

post-functionalization and molecular-weight control are important.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7666950B2 focuses on direct hydrogenation of NBR in a spinning disk reactor, while EP2473280B1 and US9150669B2 use simultaneous hydrogenation/metathesis to lower molecular weight during HNBR formation.
- Catalyst systems differ substantially: US7666950B2 broadly cites rhodium/ruthenium/titanium and other metals; EP2473280B1 and US9150669B2 center on ruthenium/osmium metathesis-type catalysts; the precursor patents EP2478021B1, EP2554558A1, EP2623518A1, and US20140357801A1 rely on radical/RAFT polymerization rather than hydrogenation catalysts.
- Hydrogenation-process patents emphasize residual double bond reduction and viscosity control, with reported HNBR RDB values dropping to near zero in EP2473280B1 and US9150669B2; precursor patents instead emphasize monomer ratios, solvent systems, conversion, and molecular-weight control before hydrogenation.
- EP2478021B1 and EP2554558A1 both use acrylonitrile/butadiene solution polymerization, but EP2478021B1 highlights modifier-controlled synthesis in DMAc, whereas EP2554558A1 emphasizes mixed-solvent systems with a main solvent fraction of 70 to 99.9 vol%.
- EP2623518A1 and US20140357801A1 introduce functional RAFT chain-transfer agents to create alkyne- or tetrazole-functional NBRs, enabling post-polymer coupling; these are not direct HNBR synthesis routes but are relevant to precursor architecture and downstream property tuning.
- Across the hydrogenation patents, lower catalyst loading and shorter reaction times are associated with lower residual double bond content and lower or moderate Mooney viscosity, while higher or less selective catalyst systems can leave higher RDB and higher viscosity.
- Coagulation/isolation methods are explicitly described in the precursor patents as ethanol precipitation or solvent removal, whereas the hydrogenation patents focus more on reactor configuration and catalyst performance than on isolation chemistry.

4. CONCLUSION

The patent set shows that HNBR synthesis is best approached through either optimized direct hydrogenation of NBR or through precursor engineering followed by hydrogenation. The strongest direct HNBR evidence supports ruthenium-based or related homogeneous catalyst systems, with process intensification in a spinning disk reactor and hydrogen pressures in the 3.5 to 100 bar range. For simultaneous hydrogenation/metathesis routes, the evidence favors ruthenium/osmium catalysts that can deliver low residual double bond content and molecular weights in the 20,000 to 250,000 range with Mooney viscosity between 1 and 50. For precursor synthesis, solution

polymerization in organic solvents, often with RAFT or modifier control, is repeatedly used to tune molecular weight and functionality before hydrogenation. Based on the evidence, the most defensible synthesis strategy for HNBR is controlled NBR precursor preparation with acrylonitrile/butadiene compositions in the typical 15 to 40 wt% nitrile and 60 to 85 wt% diene range, followed by hydrogenation using a selective homogeneous catalyst system under conditions that minimize viscosity rise and preserve molecular-weight distribution.

5. REFERENCES

- US7666950B2 | Process for preparing hydrogenated nitrile rubbers | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7666950B2/en>
- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>
- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
- EP2623518A1 | Nitrile rubbers coupled with bistriazolyl groups, production and use of same | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2623518A1/en>
- EP2554558A1 | Method for producing nitrile rubbers in organic solvents | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2554558A1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US20140357801A1 | Nitrile rubbers coupled via bisdihydropyrazole groups, production thereof and ... | Not disclosed | US | 2014 | <https://patents.google.com/patent/US20140357801A1/en>